Autonomous Robotics & Intelligent Systems
Advanced Engineering Curriculum · Acadion Learning

Introduction & Foundations
- Autonomous Agent: Self-governing entity operating in dynamic environments
- Sensor Fusion: Combining IMU, LiDAR, and camera telemetry
- Actuation: Closed-loop motor control and torque regulation

Robotics Milestones
- 1961: Unimate becomes the first industrial robot deployed at GM
- 1997: Sojourner rover navigates the Martian surface autonomously
- 2015: Deep reinforcement learning enables agile legged locomotion
- 2024: Foundation robotics models generalize zero-shot manipulation

Autonomous Navigation Pipeline
- Environmental Sensing: Capturing point clouds and RGB visual frames
- State Estimation: Extended Kalman Filtering and SLAM mapping
- Path Planning: Global Dijkstra search and local trajectory rollout
- Motion Execution: Model Predictive Control trajectory tracking

Summary & Future Outlook
- Physical Safety: Hardware interlocks and formal verification bounds
- Human-Robot Collaboration: Compliant actuation and intent prediction
- Scalability: Fleet learning across distributed robotic deployments